Testing a sparse differential norm filter

November 13, 2026. Unsigned producer evidence [pcc:unverified].

Algorithmic proposal. Replace matching $N = cG$, where $N = A^2 - fB^2$, by the sparse derivative/convolution equation

$$W_G(N) := GN' - G'N = 0.$$

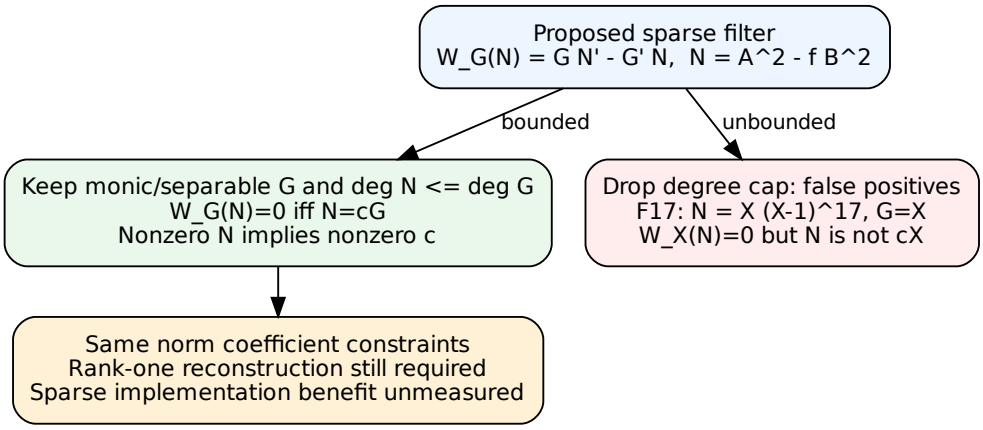
The test asks whether this adds useful constraints and whether finite characteristic permits false positives. This is an exact algebraic test, not a timing or empirical scaling campaign.

Exact result. Over any field, suppose G is monic, $\gcd(G, G') = 1$ and $\deg N \leq \deg G$. Then

$$W_G(N) = 0 \quad \Longleftrightarrow \quad N = cG \text{ for some scalar } c.$$

If $N \neq 0$, then $c \neq 0$. The theorem applies to the actual polynomial norm lift with the stated degree cap. Products of distinct node factors $X - x_i$ satisfy monicity and coprimality; the usual norm pole-degree inequalities supply the cap. No assumption $p > \deg G$ is needed.

Proof mechanism. The filter equation implies $G \mid G'N$. Coprimality gives $G \mid N$; the degree cap and monicity force a scalar quotient. Conversely, differentiating cG gives the zero filter.



Finite-characteristic check and research disposition

False positives without the cap. Over $\mathbb{F}_{17}$, $W_G(GH^{17}) = 0$ for every G, H, since $(H^{17})' = 0$. The explicit nonzero polynomial $N = X(X-1)^{17}$ passes W_X but is not cX: evaluation at1 would give $c = 0$, while evaluation at2 contradicts that value. Lean checks both the passing filter and non-proportionality. This fixture tests the polynomial filter, not an ECDLP target solve.

What was learned. Within the genuine degree window this is exactly the original scalar norm condition. It is not stronger pruning and does not remove rank one. Outside the window it can accept incorrect candidates. Kernel equivalence alone does not rule out a sparse implementation benefit; none was measured. In particular,

$$(A^2 - fB^2)' = 2AA' - f'B^2 - 2fBB'$$

still depends quadratically on the unknown coefficients. A linear routine returning $N = cG$ has not constructed A, B.

Formal map. In module `NormWronskian`, the kernel theorem is `filterZeroIff` and the actual-lift corollary is `liftedFilterIff`. The README maps all ten statements to their exact hypotheses and canonical unsigned bindings in the evidence directory.

Evidence. Source commit `923a40182`; Lean4.35.0 with unchanged pinned dependencies. Default build004 passed3145 jobs. Actual tracked-source gate passed39 declarations with zero additional trusted axioms. Failed builds001–003 are preserved; they involved the finite-field instance and natural-number casts. All final printed foundations are standard. Import-inclusive replay status is retained in the evidence directory.

Native identity record `IDC1hd9711b7c88619f6b` binds the expanded residual and source hash. The existing issuer/checker accepted32 statement-derived and32 extra stream points. It does not execute Lean or establish an algorithmic bound; the universal kernel theorem and F17 implications are checked separately.

Next research target. Nonlocal rank-one reconstruction exploiting the coupled Hankel coefficient structure. No such search routine or time bound is provided here. This test does not exclude the whole norm approach.

Sources and resources. Pinned Mathlib derivative, separability, coprimality and monic-degree lemmas; recorded Galbraith norm references; the barrier map's p-adic scope. KB unavailable, so no novelty claim. The monthly isolab budget is recorded exhausted; no paid experiment, benchmark row, timing comparison or historical graph change was made.